In the Bayesian framework, probabilities represent beliefs, and beliefs are always tentative in situations when new information can become available, so a prior probability distribution is just the distribution you start with before receiving some new information. After you receive new information, such as some biographical information about the candidates, you might update your prior distribution based on that new information—this updated distribution is now called your *posterior probability distribution*. The distinction between prior and posterior distribution is contextual, since your posterior distribution will become a new prior distribution right before you receive another set of new information. Your beliefs are continually updated as a succession of prior distributions to posterior distributions (see figure 7.5), and this process is generically called *Bayesian inference*.

Figure 7.5. Bayesian inference is the process of starting with a prior distribution, receiving some new information, and using that to update the prior into a new, more informed distribution called the posterior distribution.

